

SWINBURNE UNIVERSITY OF TECHNOLOGY

DEPLOYMENT PORTFOLIO TASK 1

My Pass to Distinction Implementation and Evidence Report

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Unit: SWE40006 Software Deployment and Evolution

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Declared target: Task 1.3 - Distinction

Submission statement: I completed Tasks 1.1, 1.2, and 1.3 on Windows. I implemented the applications and WiX projects, built both MSI packages, and verified the required installation, execution, and uninstallation for Task 1.1. I then reinstalled both final packages so the applications remain available on my PC. For Distinction, I explicitly packaged three external runtime DLLs: Newtonsoft.Json, CsvHelper, and Humanizer.

1. Executive summary

In this portfolio, I completed an end-to-end Windows deployment workflow with WiX Toolset 4.0.6 and .NET 10. For Task 1.1, I packaged a Hello World console application. For Task 1.2, I designed and built HashGuard Desktop, a WinForms checksum utility. For Task 1.3, I extended the HashGuard installer to package and verify multiple external DLL dependencies. Both final MSI packages are installed on my PC, and their deployed application files are present.

Outcome: I ran the final automated verification and all 4 tests passed. I rebuilt both MSI projects with 0 warnings and 0 errors. The final Hello World and HashGuard installations completed successfully, and I confirmed that their installed files are present. HashGuard is also available from its Start Menu shortcut.

1.1 Toolchain

Component	Installed version	Role
Visual Studio	Community 2026 18.9.0	.NET desktop workload and solution/project authoring
.NET	SDK 10.0.400; Desktop Runtime 10.0.11	Build, test, publish, and execute the applications
WiX Toolset	CLI and SDK 4.0.6	Compile the two x64 MSI packages
HeatWave	Visual Studio extension installed	WiX project templates and Visual Studio integration

1.2 Reproducibility controls

I pinned the WiX SDK and extension versions in the .wixproj files and locked the NuGet dependency versions. I also used stable product and upgrade GUIDs. I retained my final test, build, installation, execution, and metadata logs in the evidence directory, and I recorded both MSI SHA-256 hashes in evidence/33-msi-metadata.txt.

2. Task 1.1 - Pass: Hello World MSI

For the Pass task, I created a small .NET console program that prints an assignment-specific banner. I then authored a WiX x64 package that installs the published executable and its three companion runtime files to %LOCALAPPDATA%\Programs\SWE40006 Hello World. I installed the MSI silently, ran the installed executable from that directory, and captured the output. I also verified that the package could be uninstalled as required, then reinstalled the final MSI so Hello World remains on my PC.

MSI	SWE40006-HelloWorld-1.0.0-x64.msi
Product version	1.0.0
Product code	{40C76FDB-606D-4DE6-9B17-F06149341B0A}
Install result	Exit code 0; four application files present
Execution result	Installed application printed the expected Task 1.1 banner
Final state	Package reinstalled; four application files remain present

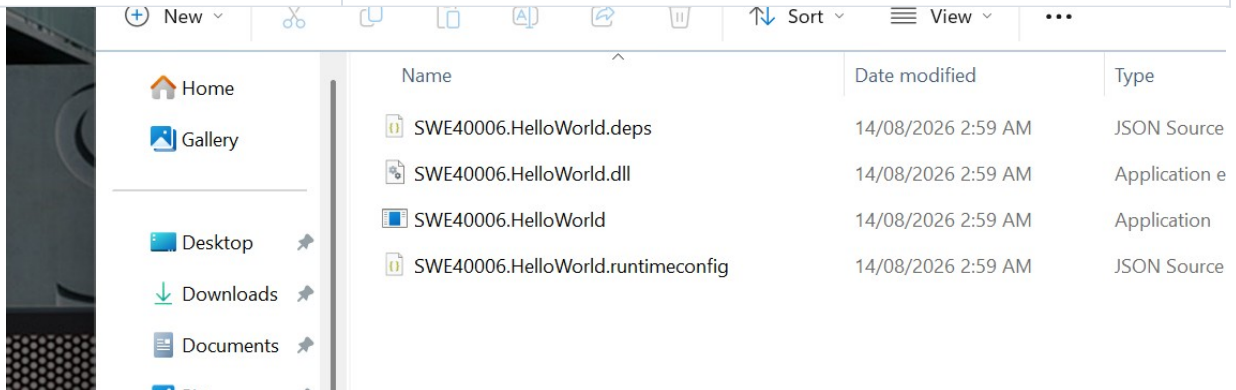


Figure 1. Task 1.1 installed directory containing the published Hello World executable and runtime files.

```
=====
SWE40006 Deployment Task 1.1
Hello World application is running.
=====
Executed successfully at 2026-08-14 02:59:41 +10:00
```

Figure 2. Console output from the installed Hello World executable.

3. Task 1.2 - Credit: HashGuard Desktop

For the Credit task, I designed and built HashGuard Desktop as an original C# WinForms application for calculating SHA-256 and SHA-512 file checksums. I implemented asynchronous hashing, optional comparison with an expected hash, a local JSON audit history, and CSV export. I also made the external dependencies visible in the interface and displayed the complete hash values for copying and verification.

3.1 Application design

Layer	Implementation	Responsibility
Presentation	MainForm.cs	File selection, progress, hash display, comparison, history, and export actions
Hashing	HashService.cs	Buffered asynchronous SHA-256/SHA-512 computation using IncrementalHash
Persistence	HistoryStore.cs	Auditable local JSON history using Newtonsoft.Json
Export	CsvExportService.cs	CSV output through CsvHelper
Presentation helper	Humanizer	Readable dates and display text

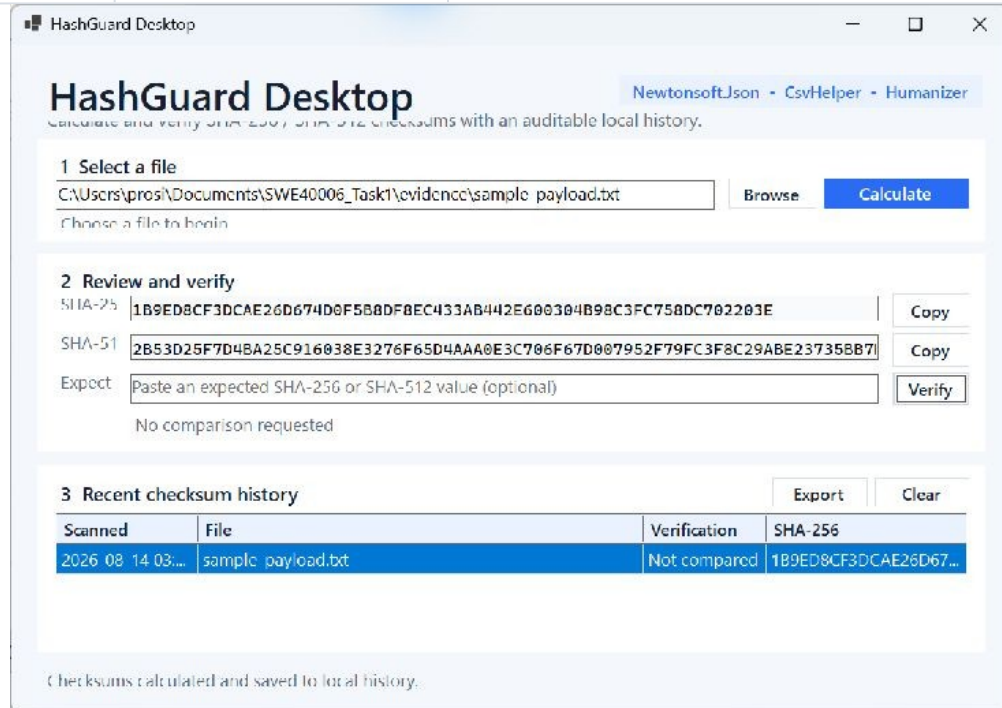


Figure 3. I ran the installed HashGuard application, calculated SHA-256 and SHA-512 values, and stored a history record.

3.2 Core checksum implementation

C# excerpt - HashService.ComputeAsync

```
using var sha256 = IncrementalHash.CreateHash(HashAlgorithmName.SHA256);
using var sha512 = IncrementalHash.CreateHash(HashAlgorithmName.SHA512);
await using var stream = new FileStream(
    filePath,
    FileMode.Open,
    FileAccess.Read,
    FileShare.Read,
    bufferSize,
    FileOptions.Asynchronous | FileOptions.SequentialScan);

var buffer = new byte[bufferSize];
long totalRead = 0;
int bytesRead;

while ((bytesRead = await stream.ReadAsync(buffer, cancellationToken)) > 0)
{
    sha256.AppendData(buffer, 0, bytesRead);
    sha512.AppendData(buffer, 0, bytesRead);
    totalRead += bytesRead;
    progress?.Report(fileInfo.Length == 0 ? 100 : (int)(totalRead * 100 / fileInfo.Length));
}

progress?.Report(100);
return new HashResult(
    fileInfo.FullName,
    fileInfo.Length,
    Convert.ToHexString(sha256.GetHashAndReset()),
    Convert.ToHexString(sha512.GetHashAndReset()));
```

I implemented HashService to stream each file in 1 MiB buffers and append every buffer to both hash algorithms. This avoids loading the entire file into memory and allows the interface to display progress for large inputs.

3.3 Automated verification

```
HashGuard.Desktop -> C:\Users\prosi\Documents\SWE40006_Task1\src\HashGuard.Desktop\bin\Release\net10.0-windows\HashGuard.Desktop.dll
HashGuard.Tests -> C:\Users\prosi\Documents\SWE40006_Task1\src\HashGuard.Tests\bin\Release\net10.0-windows\HashGuard.Tests.dll
Test run for C:\Users\prosi\Documents\SWE40006_Task1\src\HashGuard.Tests\bin\Release\net10.0-windows\HashGuard.Tests.dll (.NETCoreApp,Version=v10.0)
A total of 1 test files matched the specified pattern.

Passed! - Failed:    0, Passed:    4, Skipped:    0, Total:    4, Duration: 68 ms - HashGuard.Tests.dll (net10.0)
```

Figure 4. Final test run: 4 passed, 0 failed, 0 skipped.

Test	Expected behaviour	Result
Known hashes	SHA-256 and SHA-512 match known values	Passed
Mismatch verification	Malformed or different expected hash is rejected	Passed
JSON round trip	Newtonsoft.Json persists and restores a history record	Passed
CSV export	CsvHelper writes an export containing the history data	Passed

4. Task 1.3 - Distinction: external dependencies

Distinction criterion: To satisfy the Distinction criterion, I authored multiple runtime dependencies as individual WiX components. I verified that the final HashGuard installation contains Newtonsoft.Json.dll, CsvHelper.dll, and Humanizer.dll alongside the application.

4.1 Runtime dependency set

Dependency	Pinned version	Use in HashGuard
Newtonsoft.Json	13.0.4	Serializes and deserializes checksum history
CsvHelper	33.1.0	Exports the audit history to CSV
Humanizer.Core	3.0.10	Formats readable time and display values

4.2 WiX authoring

WiX v4 excerpt - explicit external DLL components

```
<Component Id="NewtonsoftJsonComponent" Guid="{9AAAABB3-2199-4AFB-B3F9-14AAAB88A66A}">
  <File Id="Newtonsoft.JsonDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
  x64\publish\Newtonsoft.Json.dll" KeyPath="yes" />
</Component>
<Component Id="CsvHelperComponent" Guid="{F6EBD07F-2291-4527-9317-4739766DD50B}">
  <File Id="CsvHelperDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
  x64\publish\CsvHelper.dll" KeyPath="yes" />
</Component>
<Component Id="HumanizerComponent" Guid="{EA9E4267-5AA4-4E47-9AAA-463335F372FD}">
  <File Id="HumanizerDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
  x64\publish\Humanizer.dll" KeyPath="yes" />
</Component>
```

I assigned every external file its own stable component GUID and KeyPath. In the same package, I defined the application executable, application DLL, dependency manifest, runtime configuration, installation directory, and Start Menu shortcut.

```
<Directory Id="APPLICATIONPROGRAMSFOLDER" Name="HashGuard.Desktop" />
</StandardDirectory>
</Fragment>
<Fragment>
  <ComponentGroup Id="HashGuardComponents" Directory="INSTALLFOLDER">
    <Component Id="HashGuardExeComponent" Guid="{F48AB25E-434E-4095-8F37-5B12ABFC00AC}">
      <File Id="HashGuardExe" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-x64\publish\HashGuard.Desktop.exe" KeyPath="yes" />
    </Component>
    <Component Id="HashGuardDllComponent" Guid="{815918D3-C280-438F-8553-FC98BA71E1DC}">
      <File Id="HashGuardDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-x64\publish\HashGuard.Desktop.dll" KeyPath="yes" />
    </Component>
    <Component Id="HashGuardDepsComponent" Guid="{D47537ED-5827-4176-8B14-6882DEFAC95}">
      <File Id="HashGuardDeps" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-x64\publish\HashGuard.Desktop.deps.json" KeyPath="yes" />
    </Component>
    <Component Id="HashGuardRuntimeComponent" Guid="{95DFE4D7-5DAB-418B-A235-75336D1E578F}">
      <File Id="HashGuardRuntime" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-x64\publish\HashGuard.Desktop.runtimeconfig.json" KeyPath="yes" />
    </Component>
    <Component Id="NewtonsoftJsonComponent" Guid="{9AAAABB3-2199-4AFB-B3F9-14AAAB88A66A}">
      <File Id="Newtonsoft.JsonDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-x64\publish\Newtonsoft.Json.dll" KeyPath="yes" />
    </Component>
    <Component Id="CsvHelperComponent" Guid="{F6EBD07F-2291-4527-9317-4739766DD50B}">
      <File Id="CsvHelperDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-x64\publish\CsvHelper.dll" KeyPath="yes" />
    </Component>
    <Component Id="HumanizerComponent" Guid="{EA9E4267-5AA4-4E47-9AAA-463335F372FD}">
      <File Id="HumanizerDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-x64\publish\Humanizer.dll" KeyPath="yes" />
    </Component>
  </ComponentGroup>
</Fragment>
<Fragment>
  <Component Id="HashGuardShortcutComponent" Directory="APPLICATIONPROGRAMSFOLDER" Guid="{A9A9A9A9-A9A9-A9A9-A9A9-A9A9A9A9A9A9}">
```

Figure 5. Rendered view of my authored Package.wxs, including the three external DLL components.

4.3 Build and installed-file evidence

```
HashGuardInstaller -> C:\Users\prosi\Documents\SWE40006_Task1\installer\HashGuardInstaller\bin\Release\HashGuard-Desktop-1.0.0-x64.msi

Build succeeded.
    0 Warning(s)
    0 Error(s)

Time Elapsed 00:00:00.27
```

Figure 6. Final HashGuard installer build: succeeded with 0 warnings and 0 errors.

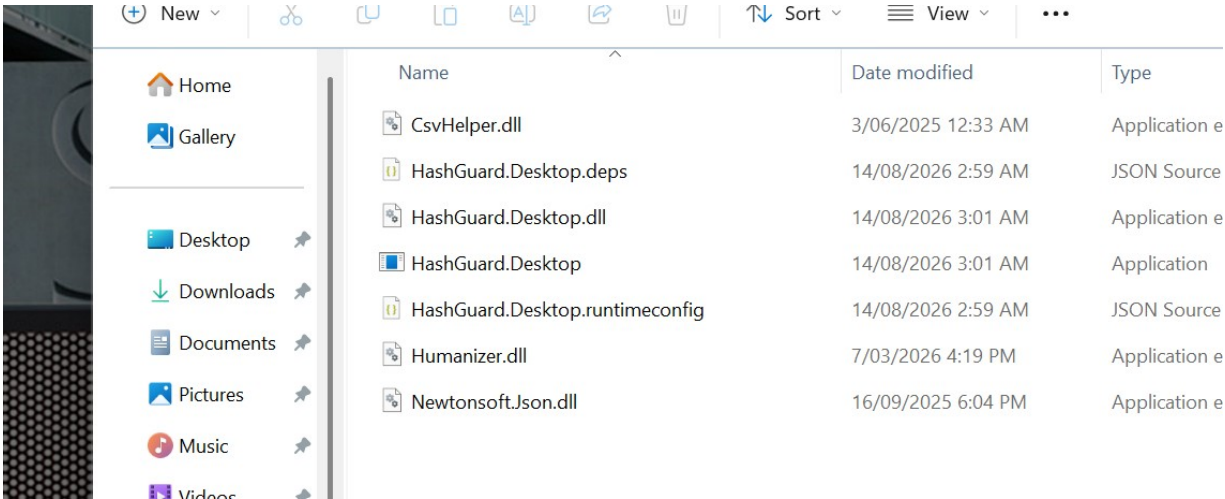


Figure 7. Installed HashGuard payload showing the application plus CsvHelper.dll, Humanizer.dll, and Newtonsoft.Json.dll.

4.4 MSI identity and installation

MSI	HashGuard-Desktop-1.0.0-x64.msi
Product code	{C57B16A8-E9E5-4EE9-8020-EE069EB409A7}
Upgrade code	{23AB1A31-A44A-41FE-BBDA-5245030BAD76}
Size	1,093,632 bytes
SHA-256	8B44ED575B560DD0C996AEB5CC1CF06C06DE8C9A3F31D669A3C50134AB488ED1
Final install	Exit code 0; 7 files plus Start Menu shortcut
Execution	Installed application launched and processed evidence/sample-payload.txt

5. Verification and evidence matrix

Requirement	Evidence	Verified result
Toolchain	27-toolchain.txt, 28-dotnet-tools.txt, 29-visual-studio-workload.json	.NET, WiX, HeatWave/Visual Studio workload recorded
Pass build	31-final-hello-msi-build.log	0 warnings; 0 errors
Pass lifecycle	06-hello-install.log, 08-hello-execution.log, 41-hello-uninstall.log, 43-restored-hello-install.log	Required installation, execution, and uninstallation verified; final package reinstalled
Credit tests	30-final-tests.log	4 passed; 0 failed
Distinction build	32-final-hashguard-msi-build.log	0 warnings; 0 errors
Distinction install	44-restored-hashguard-install.log, 20-hashguard-final-files.txt	Exit code 0; 3 external DLLs present in final installation
MSI metadata	33-msi-metadata.txt	Product identity and SHA-256 hashes recorded

5.1 Manual checksum result

Input	evidence/sample-payload.txt
SHA-256	1B9ED8CF3DCAE26D674D0F5B8DF8EC433AB442E600304B98C3FC758DC702203E
SHA-512	2B53D25F7D4BA25C916038E3276F65D4AAA0E3C706F67D007952F79FC3F8C29AB E23735BB7D6E1A41C3AE4A89D9EE1694A82E1894C4A9EBC3F1B121A640C47AF
History	One record written to the local JSON audit history

6. Troubleshooting and engineering decisions

6.1 WinForms designer serialization warning

During the initial compilation, I encountered WFO1000 because WinForms designer serialization interpreted public form state as design-time content. I refactored the form so its application services and state are private readonly fields initialized in the constructor. This separated runtime state from designer serialization and restored a clean build.

6.2 WiX v4 UI migration

My initial WiX build used the WiX v3 UIRef pattern and failed with WIX0094 because that identifier is protected in WiX v4. I migrated the package to the WiX v4 UI extension namespace and declared `ui:WixUI` with `WixUI_InstallDir` and `INSTALLFOLDER`. I pinned `WixToolset.UI.wixext` to version 4.0.6.

6.3 Installed-app runtime failure

When I installed my first HashGuard build, it terminated before displaying its window. I checked the Windows Application Event Log and identified a .NET 10 `NotSupportedException` caused by a transparent button border setting. I corrected the custom styling by explicitly setting `BorderSize` to 0, then repeated the tests, publish, MSI build, installation, and installed-application execution successfully.

6.4 MSI validation policy

On my first non-elevated WiX build, ICE validation could not run under the available Windows policy. I compensated by inspecting Windows Installer COM metadata, calculating SHA-256 hashes, performing real silent installations, checking installed files, running each application from its deployed directory, reviewing MSI logs, and confirming successful execution. My final reproducibility builds completed with 0 warnings and 0 errors.

6.5 Packaging decisions

Decision	Reason	Effect
Per-user MSI	Avoids unnecessary elevation for a student utility	Installs under LocalAppData and supports silent testing
Explicit DLL components	Makes Distinction dependency evidence unambiguous	Each external DLL has a stable component GUID and key path
Framework-dependent publish	Keeps the MSI small while targeting the installed Desktop Runtime	Runtimeconfig and deps files are deployed with the application

7. Reproduction procedure

I used the following commands from the repository root on Windows with .NET 10 and WiX Toolset 4.0.6 installed. Because I pinned the project dependencies, these commands can be rerun to reproduce my test, publish, build, and installation results.

PowerShell - build and test

```
dotnet test src/HashGuard.Tests/HashGuard.Tests.csproj -c Release
dotnet publish src/HashGuard.Desktop/HashGuard.Desktop.csproj -c Release -r win-x64 --self-contained false
dotnet build installer/HelloWorldInstaller/HelloWorldInstaller.wixproj -c Release
dotnet build installer/HashGuardInstaller/HashGuardInstaller.wixproj -c Release
```

PowerShell - install

```
msiexec /i installer\HelloWorldInstaller\bin\Release\SWE40006-HelloWorld-1.0.0-x64.msi /qn
msiexec /i installer\HashGuardInstaller\bin\Release\HashGuard-Desktop-1.0.0-x64.msi /qn
```

7.1 Expected outputs

Step	Success condition	Final observed result
Test	All automated tests pass	4/4 passed
Build	Both MSI projects succeed	0 warnings; 0 errors
Install	MSI exits 0 and files appear	Both exits 0; 4 and 7 files respectively
Run	Installed executable starts	Hello banner and HashGuard checksum result captured

8. References

FireGiant. HeatWave Community Edition and Visual Studio integration. <https://docs.firegiant.com/heatwave/>

FireGiant. WiX Toolset v4 UI extension (WixUI). <https://docs.firegiant.com/wix/tools/wixext/wixui/>

Microsoft. IncrementalHash class. <https://learn.microsoft.com/dotnet/api/system.security.cryptography.incrementalhash>

NuGet. Newtonsoft.Json 13.0.4. <https://www.nuget.org/packages/Newtonsoft.Json/13.0.4>

NuGet. CsvHelper 33.1.0. <https://www.nuget.org/packages/CsvHelper/33.1.0>

NuGet. Humanizer.Core 3.0.10. <https://www.nuget.org/packages/Humanizer.Core/3.0.10>